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In the present note we will complete the abstract theory of Section 4 of [1]. In [1, Lemma 4.2] we proved that the invariance of the discrete solution with respect to a perturbation of the data by $B(\cdot, r)$ is equivalent to the kernel inclusion $\ker \mathcal{B}_h \subseteq \ker \mathcal{B}$, and in [1, Lemma 4.4, Corollary 4.1] that the kernel inclusion implies an error estimate for u_h that does not depend on the Lagrange multiplier p. The converse of the last statement is only used informally in [1], after (3.5) and (4.28a), and we will here prove it in detail. We use the notation of [1], and equation numbers of the form (4.x) refer to [1].

We begin recalling the continuous problem: find $u \in \mathcal{V}$ and $p \in \mathcal{Q}$ such that

$$A(u,v) + B(v,p) = F(v) \;\; \forall v \in \mathcal{V}, \qquad B(u,q) = G(q) \;\; \forall q \in \mathcal{Q}, \tag{1}$$

and its conforming discretization: find $u_h \in \mathcal{V}_h$ and $p_h \in \mathcal{Q}_h$ such that

$$A(u_h,v_h)+B(v_h,p_h) = F(v_h)\,\forall v_h \in \mathcal{V}_h, \quad B(u_h,q_h) = G(q_h)\,\forall q_h \in \mathcal{Q}_h. \tag{2}$$

As in [1], both problems are assumed to be well posed for all $F \in \mathcal{V}'$ and $G \in \mathcal{Q}'$, and the discrete inf-sup condition (4.4) holds. Unlike [1], we do not assume that $A(\cdot,\cdot)$ is symmetric, and in place of (4.18) we only assume the inf-sup condition on the discrete kernel

$$\inf_{0\neq z_h \in \ker \mathcal{B}_h} \sup_{0\neq w_h \in \ker \mathcal{B}_h} \frac{A(z_h,w_h)}{\|z_h\|_{\mathcal{V}}\|w_h\|_{\mathcal{V}}} \geq \alpha_h > 0. \tag{3}$$

Since $\ker \mathcal{B}_h$ is finite dimensional, (3) is equivalent to the same condition with the arguments of A exchanged, with the same constant, and it already follows from the well-posedness of (2); we state it to give a name to α_h. Note that (4.18c) implies (3) with the same α_h. For $r \in \mathcal{Q}$ we denote by $(u_{h,r}, p_{h,r})$ the solution of (2) with F replaced by $F + B(\cdot, r)$, as in (4.6). We consider the following three properties:

(I) *invariance* (4.7): $u_{h,r} = u_h$ for all $F \in \mathcal{V}'$, $G \in \mathcal{Q}'$ and $r \in \mathcal{Q}$;

(K) *kernel inclusion*: $\ker \mathcal{B}_h \subseteq \ker \mathcal{B}$, with the kernels defined in (4.8) and (4.9);

(E) *decoupled error estimate*: there exists $C > 0$ such that for all $F \in \mathcal{V}'$ and $G \in \mathcal{Q}'$

$$\|u - u_h\|_{\mathcal{V}} \le C \inf_{v_h \in \mathcal{V}_h} \|u - v_h\|_{\mathcal{V}}. \tag{4}$$

What matters in (E) is that C does not depend on the data and that the right-hand side of (4) contains no term involving p, in contrast with the standard estimate (4.19a), which contains $\|B\|\,\alpha_h^{-1} \inf_{q_h \in \mathcal{Q}_h} \|p - q_h\|_{\mathcal{Q}}$. We claim that (I), (K) and (E) are equivalent and that, when they hold, (4) holds with $C = 2(1 + C_F)\|A\|/\alpha_h$, where C_F is the stability constant of the Fortin operator in (4.21b).

The equivalence (I)⇔(K) is [1, Lemma 4.2], whose proof only uses the well-posedness of (2). For (K)⇒(E) we cannot quote (4.25a), since the proof of [1, Lemma 4.4] uses the symmetry of A and (4.18), so we redo that step using (3). Assume (K) and let $w_h \in \mathcal{Z}_h(\mathcal{B}u)$, see (4.17). Then $u_h - w_h \in \ker \mathcal{B}_h$, and subtracting (2) from (1) gives $A(u - u_h, v_h) = 0$ for all $v_h \in \ker \mathcal{B}_h$, because $B(v_h, p_h) = 0$ and, by (K), $B(v_h, p) = 0$. Hence

$$\begin{aligned} \alpha_h \|u_h - w_h\|_{\mathcal{V}} &\le \sup_{0 \ne v_h \in \ker \mathcal{B}_h} \frac{A(u_h - w_h, v_h)}{\|v_h\|_{\mathcal{V}}} = \sup_{0 \ne v_h \in \ker \mathcal{B}_h} \frac{A(u - w_h, v_h)}{\|v_h\|_{\mathcal{V}}} \\ &\le \|A\|\,\|u - w_h\|_{\mathcal{V}}, \end{aligned}$$

and the triangle inequality, together with $\alpha_h \le \|A\|$, gives

$$\|u - u_h\|_{\mathcal{V}} \le 2\,\frac{\|A\|}{\alpha_h} \inf_{w_h \in \mathcal{Z}_h(\mathcal{B}u)} \|u - w_h\|_{\mathcal{V}}, \tag{5}$$

which is (4.20a) with $\|A\|/\alpha_h$ in place of $\sqrt{\|A\|/\alpha_h}$. Combining (5) with (4.23), whose proof only uses the Fortin operator, gives (E) with the claimed constant. If A is in addition symmetric and satisfies (4.18), the energy argument of [1, Lemma 4.4] gives the smaller constant $2(1 + C_F)\sqrt{\|A\|/\alpha_h}$ of (4.25a). It remains to show (E)⇒(I). Fix $F \in \mathcal{V}'$, $G \in \mathcal{Q}'$ and $r \in \mathcal{Q}$. By linearity of (2), $u_{h,r} - u_h$ is the first component of the discrete solution with data $\widetilde{F} = B(\cdot, r)$, $\widetilde{G} = 0$. By [1, Lemma 4.1] with $F = 0$ and $G = 0$, the exact solution of (1) with these data is $\widetilde{u} = 0$, $\widetilde{p} = r$. Since (4) holds for all data with the same constant and $0 \in \mathcal{V}_h$, we obtain

$$\|u_{h,r} - u_h\|_{\mathcal{V}} = \|\widetilde{u} - (u_{h,r} - u_h)\|_{\mathcal{V}} \le C \inf_{v_h \in \mathcal{V}_h} \|0 - v_h\|_{\mathcal{V}} = 0, \tag{6}$$

i.e. $u_{h,r} = u_h$, which is (I). This proves the claim.

We observe that the argument only uses that the right-hand side of (4) vanishes when $u = 0$. Therefore the equivalence also holds with the sharper estimate (5), whose infimum is over $\mathcal{Z}_h(\mathcal{B}u)$ defined in (4.17), since $\mathcal{Z}_h(\mathcal{B}0) = \ker \mathcal{B}_h \ni 0$. In fact (E) can be weakened to asking that $u_h = 0$ whenever $u = 0$, which, when u is the stress, is the definition of material robustness in Section 3.4 of [1].

In the examples of Section 4.5 of [1] we have $\|A\| \leq M/\mu$, $\alpha_h \geq \alpha/\mu$ and $C_F \leq \|B\|/\beta_h$ bounded uniformly in h, so this constant does not depend on μ. Hence, for the strongly symmetric schemes with $\nabla \cdot \Sigma_h^{\mathrm{sym}} = V_h$, the estimate

$$\|\sigma - \sigma_h\|_{\mathrm{div}} \leq C \inf_{\tau_h \in \Sigma_h^{\mathrm{sym}}} \|\sigma - \tau_h\|_{\mathrm{div}} \tag{7}$$

for all data, i.e. (3.5) and (4.28a), is equivalent to the kernel inclusion (4.13) and to material robustness. Conversely, for a weakly symmetric scheme violating (4.14) no such estimate holds, and a term such as $\mu \inf_{\zeta_h \in \Xi_h} \|\omega - \zeta_h\|$ in (4.29a) cannot be removed; scenario (i) of Section 4.6 of [1] makes it blow up while $\|\sigma\|_{\mathrm{div}}$ stays bounded.

References

[1] Brubeck, P., Parker, C. and Zerbinati, U., 2026. Achieving material robustness via symmetric stress finite element discretizations. Computer Methods in Applied Mechanics and Engineering, 461, p.119262.